Autonomous Smart Irrigation System Using AI — Literature Review

Papers 1–6 of 30 • A4 Landscape • 9-column matrix

S.No.	Title	Author(s) + Year	Technique / Methods	Dataset	Metrics / Error Metrics	Result	GAP / Limitation	Key Strength / Best Feature
1	Autonomous Crop Irrigation System using Artificial Intelligence	Savita Choudhary; Vipul Gaurav; Abhijeet Singh; Susmit Agarwal — 2019	Random Forest Classification/Regression for rainfall; ARIMA for rainfall time series; PLSR for crop water requirement; Raspberry Pi + NodeMCU/IoT control.	Government of India rainfall data; initial deployment focused on Punjab/wheat. Soil/weather inputs and expert irrigation decisions were also used.	Rainfall prediction accuracy; irrigation decision accuracy; train/validation/test split 70/15/15.	Controlled-environment testing reported 80% overall accuracy. Punjab annual rainfall prediction: 545 mm vs 549.5 mm, reported 83% accuracy. Pump is automatically controlled based on predicted rain and soil moisture.	Depends on weather-station information; authors propose replacing it with on-premise sensors for rural deployment. Generalization to other crops/regions requires adaptation.	Strong autonomy: predicts rainfall and water then independently decides pump ON/OFF a human intervention.
2	Smart irrigation system using IoT and machine learning methods	Authors/year not clearly stated in the uploaded PDF text	Supervised ML; decision-tree model emphasized; IoT sensing, preprocessing, API/control integration.	“Crop Irrigation Scheduling” Kaggle dataset: Crop Type, Crop Days, Soil Moisture, Temperature, Humidity, Irrigation; real-time IoT sensor data for deployment.	Accuracy, precision, recall, F1-score; decision tree described as best balanced model.	Decision tree showed the best balance of accuracy, precision, recall and F1 among the evaluated models; exact numerical scores were not clearly available in the retrieved paper text.	Exact author/year and full numerical performance are not clearly reported in the uploaded text; model comparison could be expanded with more datasets/field validation.	Combines a public irrigation-scheduling datas sensing and connects predictions directly to i
3	AI-Driven Self-Adaptive Smart Irrigation System Using IoT, Computer Vision and Machine Learning	Ramya S Yamikar; Aravinda T V; Krishnareddy K R; Ramesh B E — 2026	ESP32 + ESP32-CAM; FastAPI backend; image preprocessing (Gaussian Blur, CLAHE, HSV segmentation, contours); adaptive ML; closed-loop control.	Real-time sensor readings: soil moisture, temperature, humidity, water flow; periodic plant images. Data stored in MongoDB.	Paper emphasizes comparative/functional evaluation; no clear numerical accuracy metric was found in the retrieved result section.	Proposed system dynamically adjusts irrigation using environmental feedback and visual plant stress. Comparison table reports adaptive AI, sensor+vision, self-adaptive learning and optimized water usage versus fixed-threshold control.	Paper is primarily a framework/prototype contribution; quantitative field-performance metrics are limited compared with the strongest experimental papers.	Plant vision is the standout feature: leaf curlin modify irrigation thresholds, creating a feedba adaptive loop rather than soil-moisture-only c
4	AI-Driven Smart Irrigation System Using Edge-Based Embedded Controllers	Fahad Al-Jame; Perera Manthila — 2026	Lightweight ML/TinyML; Raspberry Pi 4 for AI inference; STM32 for actuation; local edge decision-making using environmental and soil-moisture data.	Semi-arid agricultural field-test data; local climatic/soil-moisture measurements including temperature, humidity, rain and crop-specific thresholds.	Decision accuracy, inference latency, water-use reduction, crop-yield improvement, power consumption.	Decision accuracy >94%; inference time 34 ms; water use reduced up to 38%; crop yield increased up to 20%; power consumption <2.5 W.	Model needs retraining for another crop/climate; future work proposes reinforcement learning, continual/federated learning and solar harvesting.	Excellent match to autonomous irrigation: on-device edge AI, low latency, low power and direct actuation.
5	Irrigation with Artificial Intelligence: Problems, Premises, Promises	Hanyu Wei; Wen Xu; Byeong Kang; Rowan Eisner; Albert Muleke; Daniel Rodriguez; Peter deVoil; Victor Sadras; Marta Monjardino; Matthew Tom Harrison — 2024	Review of AI/ML in irrigation; human-centred AI, explainability, transfer learning, DSS, predictive analytics and stakeholder feedback.	Literature/review corpus; not a single experimental dataset.	Review-level assessment of AI applications and barriers; no single model accuracy/error metric.	Identifies opportunities in human-centred AI, explainability and transfer learning; highlights data ownership, trust, regulatory and ethical challenges.	No single field prototype or unified benchmark because this is a review; implementation details and quantitative results vary across cited studies.	Strongest contribution is the human-centred p should incorporate farmer expertise, interpret contextual adaptation.
6	IoT Based Smart Irrigation System using Artificial Intelligence	G. Nanthakumar; K. Pazhanivel; Abinesh A; Ganesh K. R; Harish V; Nagendran S — 2024	AI-based decision logic; soil-moisture sensing; relay + pump actuation; LCD/buzzer feedback; IoT monitoring.	Real-time soil-moisture sensor data; exact dataset size/source not reported.	No clear numerical ML accuracy/error metric reported.	AI analyzes soil moisture and schedules on-demand irrigation; relay automatically activates the pump. System aims to reduce over-irrigation, waterlogging and waste.	Specific AI algorithm, dataset size and quantitative experimental metrics are not clearly reported; stronger model benchmarking is needed.	Simple end-to-end prototype with direct AI-to actuation and real-time user feedback.

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S.No.	Title	Author(s) + Year	Technique / Methods	Dataset	Metrics / Error Metrics	Result	GAP / Limitation	Key Strength / Best Feature
7	Smart Irrigation System Using Artificial Intelligence and IOT	C.G. Anu Chandra; Dinesh Kumar S; Mohammad Aousaf; Sumitra Sharma Phurailatpam — 2025	AI-powered decision support; regression/predictive analytics; weather-aware processing; cloud platform; automated valve/pump control.	Soil moisture, temperature, humidity, light intensity, flow and weather API data; historical + real-time observations.	Predictive accuracy/irrigation efficiency discussed; exact numerical model metrics not clearly reported.	Prototype maintained stable moisture predictions across soil/weather conditions; controlled evaluation reported significantly lower water use than manual irrigation, but no exact percentage was stated.	AI model details are broad (regression/neural-network options); quantitative benchmark values are limited. Current system relies strongly on alerts/manual intervention in some workflow stages.	Strong modular pipeline from sensing fi cloud fi automatic valve control, with weather integr accessible dashboard.
8	AI and IoT Based Irrigation System Using Soil Moisture	Rakhi Akhare — 2023	Not clearly stated in the supplied matrix text.	DHT11 temperature/humidity, soil-moisture sensor, LDR/light; weather/environmental data.	Paper proposes PLSR validation but does not provide a clear numerical accuracy/error result in the retrieved text.	Designed to determine optimal irrigation quantity automatically from environmental/soil factors.	Limitations stated: hardware/software cost, system complexity, sensor/data accuracy, PLSR suitability, limited user customization, and internet dependence.	Clear, lightweight PLSR-based design using practical sensors; directly targets automatic w estimation.
9	Integrating Artificial Intelligence into an Automated Irrigation System	Nicoleta Cristina Gaitan; Bianca Ioana Batinas; Calin Ursu; Filaret Niculai Crainiciuc — 2025	OpenAI models for IoT data processing/climate reports; real-time monitoring; WebSocket data transmission; predictive recommendations.	Distributed device data: temperature, humidity, soil moisture; 30-minute compressed intervals; historical 24-hour context for AI prompts.	No clear conventional ML accuracy/error metric reported.	AI generates predictive recommendations and climate analytics; system supports automated irrigation monitoring and resource optimization.	Cloud/AI dependence and prompt-based decision workflow can introduce connectivity, cost, explainability and control-safety concerns; quantitative irrigation savings are not clearly reported.	Integrates AI analytics with a scalable distribu application and interactive geographic visuali
10	Smart Irrigation Powered by Artificial Intelligence (AI)	Harini S; Shobana A; Sapna S; Vedhashree R; Prabhakaran M — 2024	AI/ML-based irrigation planning; Arduino Uno; DHT11; soil-humidity sensor; solenoid valves; Blynk IoT monitoring.	Sensor data from underground-water-well implementation; exact dataset size/source not reported.	No clear AI accuracy/error metric reported.	Authors report potential savings of up to 50% in operational costs; system supports remote monitoring and automatic water supply control.	Specific AI model and quantitative irrigation-performance validation are not clearly established; more rigorous field benchmarking is needed.	Combines low-cost embedded hardware, aut and remote Blynk monitoring with AI-based d
11	Intelligent Irrigation System With Smart Farming Using ML and Artificial Intelligence Techniques	P. Manikandan; S. Saravanan; C. Nagarajan — 2024	Ensemble SVM-KNN; IoT sensors; cloud service/data storage; web application for visualization and monitoring.	Sensor-generated agricultural data; paper states no dataset was generated/shared as a conventional public dataset.	Accuracy: SVM 72%, KNN 76%, SVM-KNN ensemble 90%.	Ensemble SVM-KNN achieved 90% accuracy, outperforming individual SVM and KNN (72% and 76%).	Dataset details/field-scale validation are limited; system combines smart farming and irrigation, so irrigation-specific validation is less detailed.	Demonstrates a measurable benefit from ens over individual classifiers and provides an int monitoring interface.
12	Design and Implementation of an Integrated IoT and Artificial Intelligence System for Smart Irrigation Management	Taoufik Benhmad; Chibani Belgacem Rhaimi; Saleh Alomari; Leenah Aljuhani — 2024	AI prediction/regression; IoT sensor network; cloud-based wireless communication; remote monitoring/control.	Real-time environmental data for date palms in southern Tunisia/Sahara oasis; sensor-network data.	Water-use reduction, yield/crop-health outcomes; exact model accuracy not clearly stated in retrieved text.	Reported substantial water reduction, improved crop health and yield; the paper reports up to 35% water reduction and 25% yield increase in its results discussion.	Application is focused on date palms/Sahara conditions; broader crop/generalization and robustness across heterogeneous environments need further validation.	Strong real-world context: targets water scarc oases and combines AI prediction with remot management.

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S.No.	Title	Author(s) + Year	Technique / Methods	Dataset	Metrics / Error Metrics	Result	GAP / Limitation	Key Strength / Best Feature
13	AI-Driven Smart Irrigation and Resource Optimization for Sustainable Precision Agriculture	Shambhavi Kaushik; Karan Singh — 2025	AI + IoT; machine-learning predictive model; cloud analytics; adaptive irrigation scheduling; optimization module; edge/cloud pipeline.	Real-time soil moisture, weather forecasts and crop-specific parameters; case studies across diverse agro-climatic regions.	Water-use efficiency, crop yield, agrochemical/resource consumption; detailed model-specific accuracy not clearly reported.	Case studies report improved water-use efficiency, reduced agrochemical consumption and enhanced crop yield.	Paper describes a broad framework/case-study approach; robustness to sensor failure and heterogeneous fields remains an identified gap.	Combines irrigation with broader resource op than only pump ON/OFF decisions.
14	Sustainable agriculture through IoT-driven smart irrigation with explainable AI	S. G. Rahul; S. J. Rithikkaa; T. M. Amirthalakshmi; M. Gayathri — 2026	Isolation Forest anomaly detection; five ML classifiers; XGBoost; SHAP; two-task architecture for irrigation classification + pump runtime regression; IoT/NodeMCU.	Real field prototype in Ponneri, Tamil Nadu; soil moisture, temperature, humidity, light and rainfall every 15 min for 90 days; 25,920 raw points; 5,000 hourly ML-ready records.	AUC-ROC, F1, inference latency, training time, pipeline latency, water consumption.	XGBoost AUC-ROC 0.995; F1 0.992 ± 0.003; training 2.40 s; inference 0.004 ms/sample; sensor-to-actuator latency ~4–6 s; water use reduced 35.1%.	Outdoor sensor degradation and model trust are key issues; long-term multi-crop/general-region validation remains important.	Exceptional combination of real hardware dat filtering, explainability and real-time actuation to a modern autonomous system.
15	A Smart IoT-based Irrigation System Design using AI and Prediction Model	Faeze Behzadipour; Mahmod Ghasemi Nezhad Raeini; Saman Abdanan Mehdizadeh; Morteza Taki; Bijan Khalil Moghadam; Mohammad Reza Zare Bavani; Jaime Lloret — 2023	Regression model in SPSS; Genetic Programming in MATLAB; combined sensor + RGB plant-image model; microcontroller-based intelligent irrigation.	35 m² greenhouse in Khuzestan, Iran; 12 cucumber plants; 16 capacitive soil-moisture sensors; soil/environmental sensors + RGB leaf images.	R^2 and standard error; water saving; yield/color indicators.	Combined sensor+image Genetic Programming: R^2 = 0.88, standard error = 0.03; regression baseline R^2 = 0.86, SE = 0.21; intelligent irrigation saved 11% water with no significant yield/color change at 5% level.	Tested on cucumber/greenhouse setting; authors suggest future adaptation to other crops and adding nutritional-deficiency detection.	Multimodal sensing is the key strength: comb soil/environmental measurements with plant before controlling irrigation.
16	Implementation of Smart Irrigation using IoT and Artificial Intelligence	Tace Y.; Elfilali S.; Tabaa M.; Leghris C. — 2023	KNN, SVM and Naive Bayes; Arduino sensor acquisition; Node-RED monitoring/storage; MongoDB Cloud; supervised learning with pump-decision labels.	Soil humidity, air humidity, temperature and rain sensors; controlled-environment plant data.	Accuracy and RMSE; KNN accuracy 98.4%, RMSE 0.016.	KNN achieved 98.4% accuracy and 0.016 RMSE among the compared models.	Authors note need for larger databases and more algorithms; semi-supervised learning is proposed as future work.	Directly maps sensor values to pump decisio compares multiple ML models in a simple dep workflow.
17	Development of IoT and AI based Smart Irrigation System	Yunus Emre Kunt — 2025	LSTM for irrigation-need prediction; ESP32; Blynk; Google Sheets; rain/soil moisture/DHT11/water-flow sensors; relay + solenoid valves.	Historical sensor data collected by prototype; exact dataset size and public source not clearly reported in retrieved text.	System testing/accuracy discussed, but no clear numerical LSTM performance metric was visible in retrieved sections.	System supports AI-driven, automatic and manual modes; fault detection is included to prevent over-irrigation/hardware failure.	Prototype is mainly a system-development study; detailed model benchmarking and large-scale field validation are limited.	Very close to the target project architecture: edge controller + sensors + valves + automatic control.
18	Smart Irrigation Systems Using IoT and AI for Efficient Water Resource Management	Aakansha Soy; Mrunal Salwadkar — 2024	Random Forest Regression (RFR); LSTM; Q-Learning reinforcement learning; IoT/WSN; edge computing; automated pumps/valves.	Real-time soil moisture, temperature, humidity and environmental sensor data; tomato and maize test plots are discussed.	R^2 for RFR; RMSE for LSTM; over-irrigation event reduction; water saving; yield improvement; latency/energy.	RFR: R^2 = 0.91 for short-term soil-moisture prediction; LSTM: RMSE = 0.07 for five-day soil-water dynamics; Q-Learning reduced over-irrigation events by 45%; system reports up to 38% water savings and 12–18% yield improvement; edge processing reduced transmission latency by 42% and IoT communication energy use by 18%.	Scalability across large farms, sensor calibration on heterogeneous soils, affordability for smallholders, and security/privacy remain challenges.	Strong multi-model AI approach: supervised prediction + temporal deep learning + reinforcement learning with edge computing and automatic actuation.

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S.No.	Title	Author(s) + Year	Technique / Methods	Dataset	Metrics / Error Metrics	Result	GAP / Limitation	Key Strength / Best Feature
19	Smart Irrigation Systems: A Comprehensive Review of IoT, AI, and Sustainable Agriculture Technologies	Naz Mohammed Ghazi; Hussain Thahir Tahir; Skala Salam Othman; Dalya Aydeen Hawar; Murad Abdullah Abdulkadir; Cem Korkmaz — 2025	PRISMA-based review of IoT/LPWAN, RF/XGB/LSTM/CNN-LSTM/Transformers, MPC, RL, fuzzy logic, anomaly detection and edge/cloud architectures.	Review corpus: >150 studies screened; representative subset of >60 high-impact sources synthesized; primary window 2017–2025.	Review-level synthesis rather than one model's accuracy; reports typical water-use reductions and system outcomes across studies.	AI/sensor-driven scheduling commonly reduces water use 15–40% while maintaining/improving yield; review identifies sensor drift, connectivity, proprietary architectures and explainability as barriers.	No single experimental system; results aggregate heterogeneous studies and should not be treated as one benchmark.	Excellent roadmap for selecting architecture, methods, edge/cloud design and identifying r for your project.
20	Automated Irrigation System Using Artificial Intelligence (AI)	Prof. Anand Tilagul; Arunkumar R; Darshan TP; Gajendra BR; Likith Gowda KM — 2024	IoT + Artificial Intelligence; Partial Least Squares Regression (PLSR); DHT-11 temperature/humidity sensor; soil-moisture sensor; LDR; automatic irrigation control. Sensor data are analyzed to determine crop water requirements and trigger irrigation.	Real-time soil moisture, temperature, humidity and light-intensity sensor data. Dataset size/source and train/test counts are not clearly reported.	No clear numerical accuracy, RMSE, R^2, precision/recall or other model-error metric is reported in the Results section.	The paper presents demo models and an automated irrigation workflow. It describes AI-based prediction and automatic control, but does not provide a clear numerical performance result in the Results section.	Data quality, scalability and cost-effective deployment are identified as challenges. Detailed quantitative field validation and dataset/model benchmarking are limited.	Best feature: a clear closed-loop workflow—s AI/PLSR → water-requirement prediction → a irrigation control—with low human interventio
21	Multi-Model AI-Based Smart Irrigation System for Precision Water Management in Agriculture	Abith M A; Priyanka D K; Vijet R; Ajay; J. Reshma Farhin — 2026	Multi-modal AIoT; LSTM for soil-moisture forecasting; weather APIs for rainfall-aware planning; crop-specific hydration/FAO Penman-Monteith models; Linear Programming and Genetic Algorithms; ESP32/Raspberry Pi; MQTT; solenoid valves/pumps; web/mobile monitoring.	90-day cropping-season evaluation on tomato and brinjal. Soil moisture, temperature, humidity, solar/rain information, weather/API data and crop-growth parameters. Exact dataset size/source is not clearly reported.	Prediction accuracy; water-consumption reduction; crop-yield improvement; system latency. Reported ≈93% prediction accuracy and ≈1.2–1.3 s latency.	Detailed case study reports ≈38% water-consumption reduction and nearly 12% yield increase. The conclusion reports water usage down almost 38%, yield improvement 10–12%, and labor needs reduced nearly 40%.	Broader validation across crops/climates, sensor failures/data noise and deployment cost remain concerns. The paper identifies scalability and real-world robustness as continuing challenges.	Best feature: true multi-modal decision makin + weather forecasting + crop-specific hydratio models—followed by optimization and autom rather than relying on soil moisture alone.
22	Internet of Things Sensors and Support Vector Machine Integrated Intelligent Irrigation System for Agriculture Industry	G. Kranthi Kumar; Manoj L. Bangare; Pushpa M. Bangare; Chanda Raj Kumar; Roop Raj; José Luis Arias-Gonzáles; Batyrkhan Omarov; Md. Solaiman Mia — 2024	IoT soil/humidity sensing; CFS feature selection; K-Means clustering; SVM; Random Forest; Naïve Bayes; hybrid K-Means + SVM.	IoT soil and humidity data plus historical soil/humidity information; the paper states that the supporting dataset is openly available through SmartFasal.	Accuracy, speci→city, recall, precision; water-saving outcome.	K-Means + SVM achieved 98.5% accuracy and outperformed Random Forest and Naïve Bayes; the paper also reports up to 20% fresh-water saving.	Requires stronger validation across broader →eld conditions, crops and deployment environments; sensor/data quality remains important.	Hybrid clustering + SVM approach combines data grouping with classi→cation for water-demand and irrigation decisions.
23	IoT with Evolutionary Algorithm Based Deep Learning for Smart Irrigation System	P. Suresh; R. H. Aswathy; Sridevi Arumugam; Amani Abdulrahman Albraikan; Fahd N. Al-Wesabi; Anwer Mustafa Hilal; Mohammad Alamgeer — 2022	IoT sensors; Arduino; DSVM-based regression; clustering; Artificial Immune Optimization Algorithm (AIOA); Deep Belief Network (DBN); cloud processing.	Soil moisture, temperature, air temperature and humidity; predicted weather/environmental variables including pressure, precipitation, solar radiation and wind speed.	Accuracy, precision, recall and F1-score; 5-fold cross-validation and train/test evaluation.	Reported average accuracy 97.1%, precision 98.0%, recall 98.1% and F1-score 98.0%; the paper reports higher performance than the compared baseline models.	Cloud dependence and computational complexity make the approach heavier than lightweight edge models; real-time/mobile deployment is suggested as future work.	Combines evolutionary optimization and deep learning with IoT data in a multi-stage irrigation decision pipeline.
24	The Application of Machine Learning Techniques for Smart Irrigation Systems: A Systematic Literature Review	Abiadi Younes; Zouhair Elamrani Abou Elassad; Othmane El Meslouhi; Dauha Elamrani Abou Elassad; Ed-dahbi Abdel Majid — 2024	Systematic literature review of ML-based smart irrigation; comparison of commonly used supervised and predictive ML approaches.	55 selected studies published from 2017–2023; literature corpus rather than a single experimental dataset.	Review-level comparison including accuracy, RMSE, R^2, MAE, MSE, precision, recall, F1, NSE and related metrics used across studies.	The review concludes that ML methods can outperform conventional approaches, but well-formed, reproducible and generalizable results remain limited.	Limited generalization, inconsistent experimental settings and the need for stronger empirical validation across irrigation contexts.	Provides a systematic evidence base for selecting ML models, metrics and evaluation strategies for smart irrigation.

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S.No.	Title	Author(s) + Year	Technique / Methods	Dataset	Metrics / Error Metrics	Result	GAP / Limitation	Key Strength / Best Feature
25	Smart Drip Irrigation Systems Using IoT: A Review of Architectures, Machine Learning Models, and Emerging Trends	Nikita Jaiswal; T. Vijay Kumar; Chitra Shukla — 2025	Review of IoT drip irrigation architectures, sensors, ML/DL models, edge/cloud computing, renewable energy, mobile/web interfaces and automation.	56 peer-reviewed studies published from 2015–2024; literature corpus.	Review-level synthesis of water ef→ciency, accuracy, latency, crop productivity, energy use and adoption factors.	The review reports that many smart-irrigation studies achieve high prediction accuracy and substantial water savings, while also emphasizing scalability and reproducibility limitations.	Lack of standardized benchmarks/open datasets; interoperability, cybersecurity, real-time adaptation, long-term →eld validation and cost-bene→t analysis remain important gaps.	Excellent architecture-level roadmap for combining sensors, AI, edge/cloud processing and automated irrigation.
26	Smart Irrigation Systems in Agriculture: A Systematic Review	David Vallejo-Gómez; Marisol Osorio; Carlos A. Hincapié — 2023	Modi→ed PRISMA 2020 systematic review covering AI, ML, IoT, fuzzy logic and smart irrigation approaches in rural and urban agriculture.	170 screened articles, with 50 sources selected for detailed analysis; literature corpus.	Review-level comparison of AI/ML techniques, sensing, automation and irrigation outcomes rather than one experimental model.	Maps the state of smart irrigation research and identi→es technologies that can support automated irrigation in both rural and urban agriculture.	The review itself does not provide one uni→ed benchmark; practical deployment still depends on crop, environment, sensing and system architecture.	Broad systematic map of AI-enabled smart irrigation and its applicability to different agricultural settings.
27	Smart Irrigation System Based on IoT and Machine Learning	Y. Tace; M. Tabaa; S. El→lali; C. Leghris; H. Bensag; E. Renault — 2022	Soil-moisture, temperature, air-humidity and rain sensing; Node-RED; MongoDB; KNN; Neural Network; Naïve Bayes; SVM; Logistic Regression; web application.	Sensor data collected over months from an environment designed to support plant growth; soil moisture, temperature, humidity and rain/pump observations.	Accuracy and RMSE.	KNN achieved 98.3% accuracy with RMSE 0.12, ahead of the evaluated Neural Network, Naïve Bayes, SVM and Logistic Regression models in the reported experiment.	The authors point to the need for expanded data and further algorithms; the dataset is not a broadly reusable benchmark in the paper.	Simple, practical mapping from environmental sensor data to pump ON/OFF decisions with a web-based monitoring layer.
28	Deep Learning for Intelligent Irrigation Decision-Making: A Review	Jingxin Yu; Qinglin Qu; Shuyi Peng; Xiaoming Wei; Yinkun Li; Congcong Sun — 2025	Review of CNN, RNN, LSTM/BiLSTM, GRU, deep reinforcement learning/DQN, multimodal data fusion, hybrid models and edge-cloud architectures.	Research literature on intelligent irrigation; not a single experimental dataset.	Review-level synthesis covering accuracy, R^2, water savings, resource ef→ciency, latency, stability and related decision-optimization measures.	The review highlights model–data alignment: CNN for spatial data, recurrent models for time series, and DRL for scheduling. It reports BiLSTM soil-water-content R^2 values of 0.818–0.991 in cited work and a DQN case with 23 mm water savings.	Data quality, model generalization, computation, cost, farmer acceptance, regional adaptation and climate change response remain major challenges.	Useful framework for choosing an AI architecture according to irrigation data type and deployment constraints.
29	A Smart Irrigation Control System That Leverages Machine Learning	M. Karthikaa; J. Steffib — 2025	IoT temperature + soil-moisture sensors with ESP32; Random Forest classifier for wet/dry soil-status prediction; SMOTE, ADASYN and Mix-up for class-imbalance handling; cross-validation; automatic pump ON/OFF control; FastAPI backend for live prediction/control.	Real-world soil dataset containing temperature (°C), moisture (%) and soil status (Wet=1 / Dry=0). Dataset naturally contained more dry than wet readings.	Accuracy, Precision, Recall and F1-score; confusion matrix and ROC curve.	Random Forest baseline: 84% accuracy, 82% precision, 80% recall, 81% F1. SMOTE: 89%, 87%, 88%, 87%. ADASYN: 91%, 90%, 89%, 89%. Mix-up: 92% accuracy, 91% precision, 92% recall, 91% F1 — best result.	The paper does not report a dataset size or broad multi-field validation in the supplied text. Future enhancements proposed include weather forecasting, LSTM/GRU time-series prediction, mobile-app integration and solar-powered operation.	Best feature: systematic treatment of class imbalance using SMOTE, ADASYN and Mix-up, with Mix-up giving the strongest reported performance (92% accuracy, 91% F1), while the prediction is connected directly to automatic pump ON/OFF control.
30	AI-Driven Smart Irrigation System Using Edge-Based Embedded Controllers	Fahad Al-Jame; Perera Manthila — 2026	Edge-based TinyML/supervised Decision Tree classifier; capacitive soil-moisture sensor, DHT22 temperature/humidity sensor and rain sensor; Raspberry Pi 4 for local AI inference; STM32F103C8T6 for real-time actuation; TensorFlow Lite quantization/deployment; Python/Scikit-learn; MQTT/LoRa/Wi-Fi communication; hybrid rule-based filtering plus AI prediction. Model outputs irrigation ON/OFF plus irrigation duration in seconds.	Real agricultural field data collected for two months on vegetable farms in semi-arid zones; more than 17,000 samples; five-minute sampling interval. Inputs include soil moisture, temperature, relative humidity and rainfall. Samples were manually labeled using agronomic expertise and irrigation programs; cleaning, outlier processing and balancing were performed.	Decision accuracy; Precision, Recall and F1-score used for model evaluation; inference latency; water consumption; crop yield; power consumption. Reported average inference time was 34 ms.	Greater than 94% decision accuracy over expert irrigation labels; up to 38% reduction in water use; 20% increase in crop yield; average inference latency ≈34 ms; total system power ≤2.5 W. Six-week field experiment compared the system with conventional timer-based irrigation.	Performance is tied to the relevance and variety of training data; the model needs retraining for new crops or climates. The paper proposes continual/federated learning, reinforcement learning, solar power and multizone irrigation as future directions.	Best feature: a genuinely autonomous, offline-capable edge architecture that separates AI inference from actuation. The Decision Tree provides both irrigation necessity (ON/OFF) and duration, while Raspberry Pi + STM32 delivers low-latency, low-power real-time control without requiring cloud connectivity.